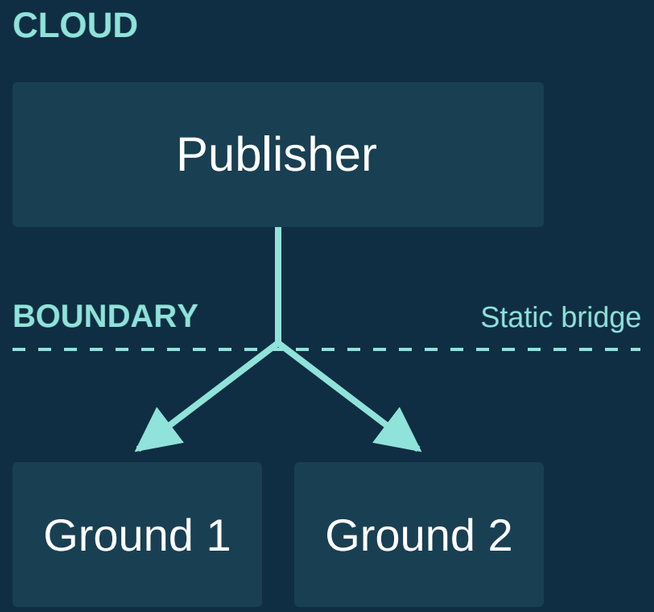
SPACE SITUATIONAL AWARENESS / EVENT INTEGRATION

Get satellite alerts to the right people on the ground.

The mission challenge: launch alerts start in a cloud environment, but the ground systems that act on them run in a separate one.

Austin Thomas

Solutions Engineer, Solace



On time
Only to authorized systems
Without custom links

WHAT SUCCESS LOOKS LIKE

What we'll show today

01

Controlled access

Only approved people and systems receive the alert.

WHAT WE WILL SHOW

Permitted consumer receives; unpermitted publish and subscribe attempts are denied.

02

Reliable delivery

Important alerts wait when an operator or system is offline.

WHAT WE WILL SHOW

A service goes down; queued alerts are delivered when it reconnects.

03

Less custom integration

New recipients can be added without building another custom connection.

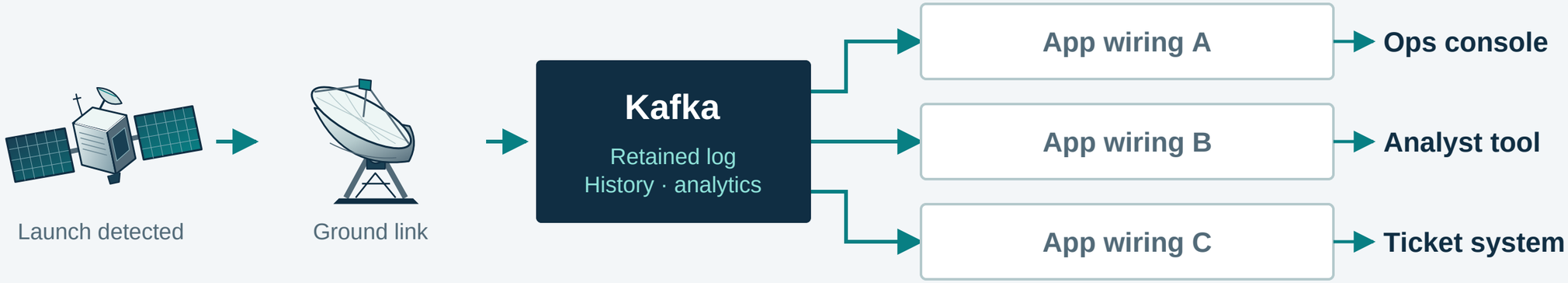
WHAT WE WILL SHOW

Add a consumer with only subscription and access changes; producer unchanged.

THE PROBLEM TODAY

Every new recipient means another custom connection.

Typical pattern



01

App wiring

Each consumer has its own custom path.

02

Alert selection

Each path has to choose which alerts it should see.

03

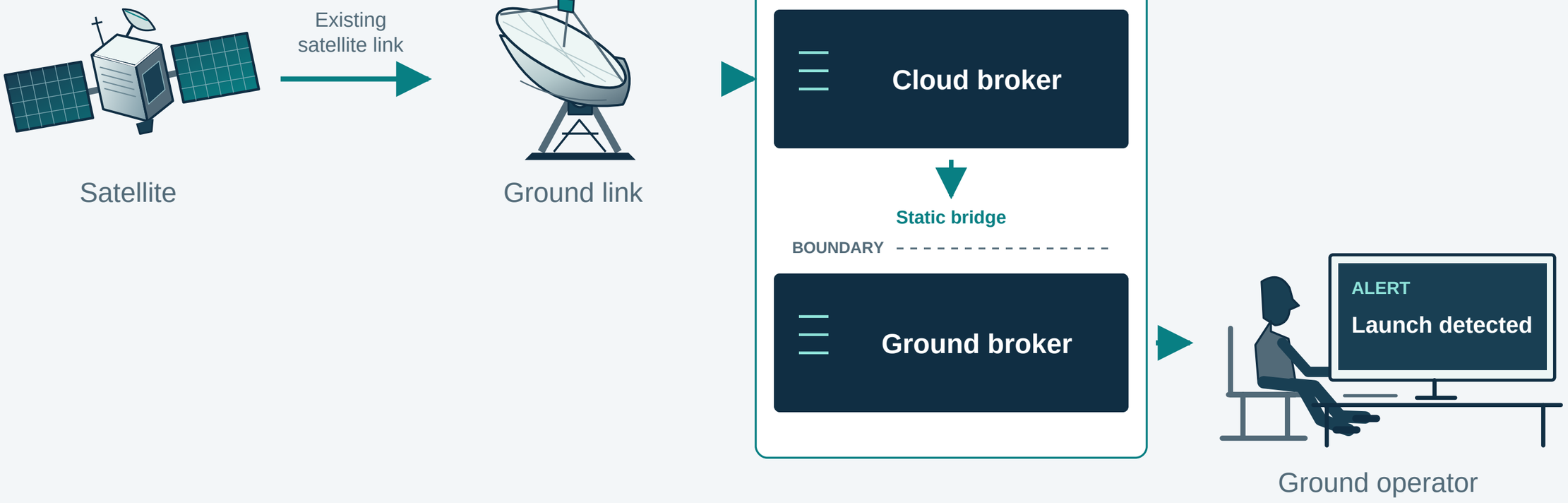
Delivery / disconnect handling

Each path has to handle an operator or system going offline.

THE SIMPLE USE CASE

A satellite has an alert. A person on the ground needs it.

Existing satellite communications stay in place.

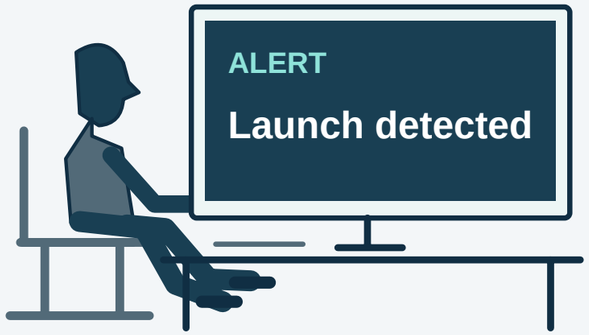


ACCESS CONTROL

The right alert reaches the right people.

The broker checks permissions on every publish and subscribe.

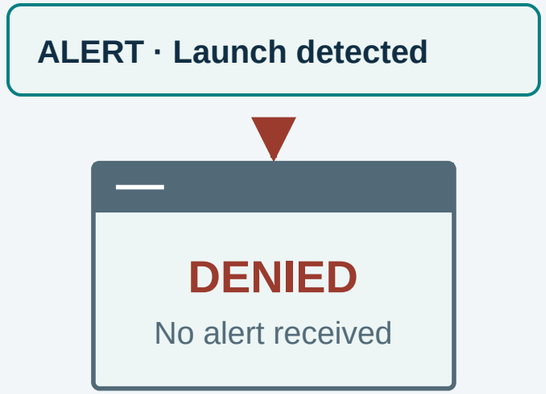
Permitted



Authorized operator

Receives the alert

Not permitted



Restricted client

Blocked by the broker

DEMO 2

GUARANTEED DELIVERY

Important alerts wait while a service is offline.

Live updates can be missed. Important alerts wait in a queue until the service reconnects.

Live updates

Direct



Service offline

Launch detected is not retained

Missed while offline

Important alerts

Guaranteed · local queue

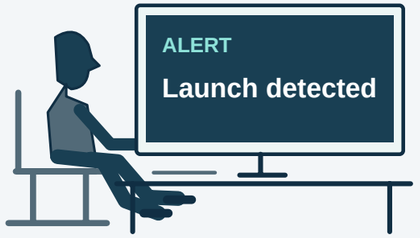


Service offline

Solace queue

Holds Launch detected

After reconnect

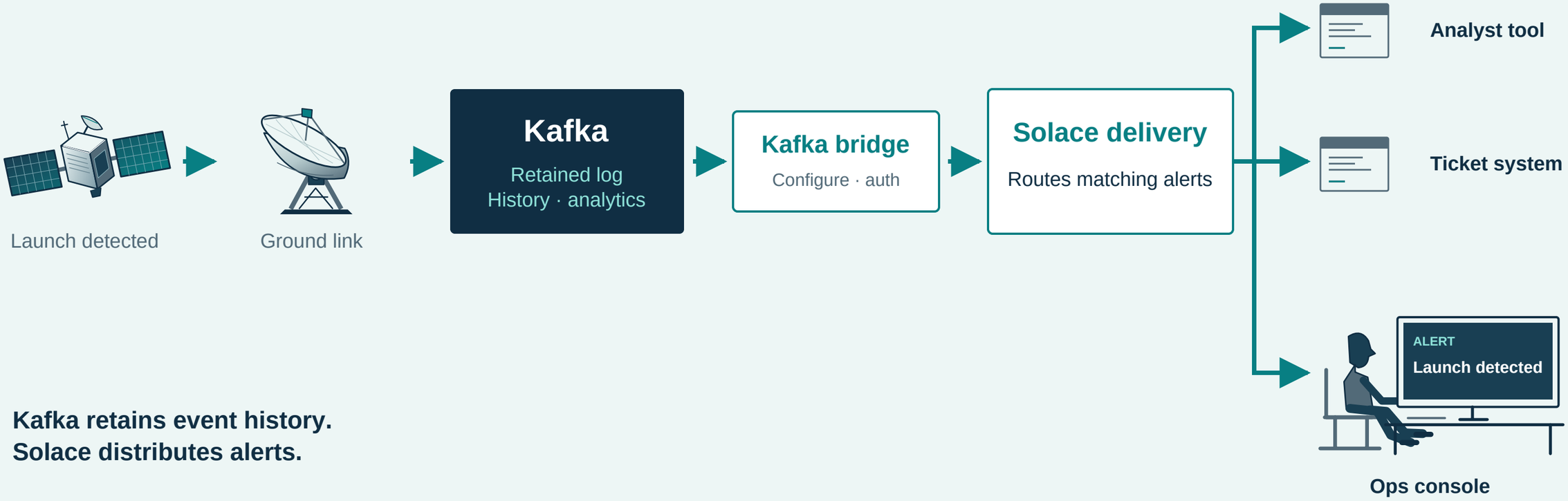


Delivered after reconnect

DEMO 3

PROPOSED PATH

Keep Kafka for history. Let Solace distribute the alerts.

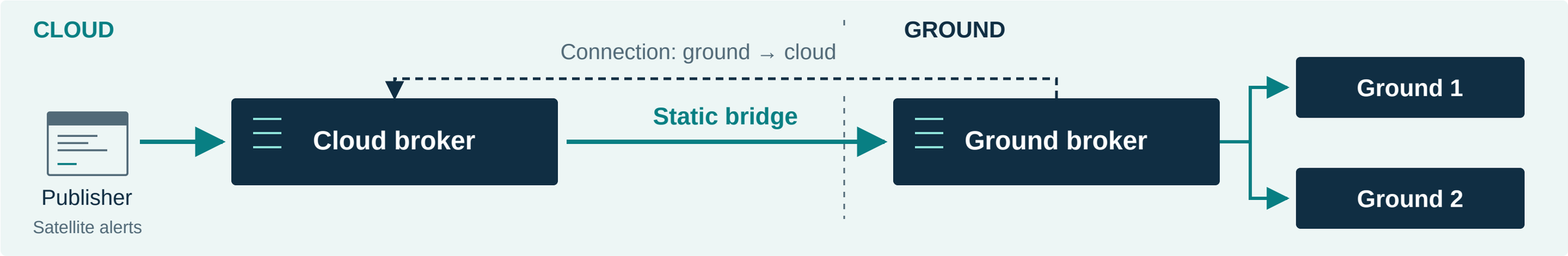


Kafka retains event history.
Solace distributes alerts.

DEMO MAP

What you will see, one step at a time

Return to this map between demos.



EXAMPLE TOPIC

Root		Classification		Site		Category		Object		Event
`ssa`	/	`unclass`	/	`ground1`	/	`launch`	/	`25544`	/	`alert`

01 Routing

Simplified routing with topic wildcards * and >

`ssa/unclass/*/launch/>`

DEMO 1

02 Access

Permitted publish and subscribe succeed; denied topics are refused by the broker.

DEMO 2

03 Delivery

A service goes offline; its alerts queue and arrive when it reconnects.

DEMO 3

BONUS · DEMO 4 Operations Watch: an AI agent explains what a service missed overnight.

De-risking Agent Mesh in federal environments

Three concerns federal teams raise, and how to address each.

Concern	Approach	How to verify
1 AI agent & data boundary AI agents and data must stay within the approved boundary.	Approved private/local model; restrict tools; inventory all outbound dependencies.	Verify whole-system data flows, unauthorized access and outbound blocking.
2 Sensitive data & official records Conversations/files may be sensitive or records with mandated retention; some require permanent archives.	Customer retention schedule; access by designation/classification; customer controls for legal holds and archival preservation.	Verify access and storage; validate retention, holds, archival preservation, backups and logs.
3 Undo & conflicting changes Can actions be reversed? What if two AI agents change the same thing at the same time?	Human-in-the-loop before irreversible actions; plan how to reverse or correct each action; coordinate conflicts.	Fully test race conditions and partial-completion failures; verify safe conflict handling, recovery and irreversible actions.

In Demo 4, the agent only observes; delivery never depends on it.

Classified use requires an approved deployment with an ATO at that level.

NEXT STEP

Prove the business value in your environment.

What the pilot covers

Use-case flow and consumers
Topics and permissions
Outage cases to test
Enclave, model, records, and recovery evidence

Staged proof of value

A short, staged pilot that proves three results.

01 **Controlled access**

Allowed and denied attempts match the permission plan

Evidence: test log and denial counts in Broker Manager

02 **Fewer missed important alerts**

Validate expected delivery in planned outage cases

Evidence: queue depth and delivery counts in Broker Manager and Insights

03 **Less custom integration effort**

Add a recipient via subscriptions and access rules

Evidence: change record, no producer changes; federal feasibility findings

Built-in monitoring in Broker Manager and Insights; can also feed an existing Grafana.

Stage 1 reuses today's demo. Later stages run in your approved environment.